

The Effects of Caffeinated Drinks on Heart Rate and Blood Pressure

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Firstly, one of the most immediate effects of caffeinated drinks on the body is a sharp rise in blood pressure. When caffeine enters the bloodstream, it blocks a natural chemical in the body that normally keeps blood vessels open and relaxed, causing them to tighten and forcing the heart to work harder to push blood through (Awashra et al., 2025). For example, Grasser et al. (2014) found that young adults who drank Red Bull showed a clear increase in both the upper and lower readings of their blood pressure compared to those who drank plain water, with the effect showing up within the same session. This rise is not simply a matter of numbers on a monitor — a higher blood pressure reading means the heart muscle itself is under greater strain every time it beats. In other words, drinking a high-dose energy drink regularly asks the heart to do more work than it should, and over time this added strain can take a lasting toll on the cardiovascular system. It is therefore clear that the blood pressure effects of caffeinated drinks, even in young and otherwise healthy people, should not be taken lightly.

Secondly, caffeinated drinks have a notable effect on how fast and how steadily the heart beats. Caffeine stimulates the nervous system, which in turn tells the heart to pump more blood per minute; at the large doses found in energy drinks, this can push the heart into an overworked state (Awashra et al., 2025). Grasser et al. (2014) observed that Red Bull consumption in young adults led to a significant increase in heart rate and in the total amount of blood the heart was pumping, adding measurably to the load the heart had to bear. More concerning still, Mandilaras et al. (2022) found that children and teenagers who consumed energy drinks experienced a rise in irregular heartbeats — extra beats that fall outside the heart's normal rhythm. Interestingly, this study also recorded a slight drop in overall heart rate, which the researchers attributed to the body trying to protect itself from a sudden pressure spike

— a reflex that itself signals the heart is under significant stress. Taken together, these findings show that caffeinated drinks can both speed up and destabilise the heart's rhythm, posing a genuine risk to young people, particularly those with existing heart conditions.

To conclude the discussion of impacts, caffeinated drinks have also been shown to reduce the flexibility of blood vessels and restrict blood flow to the brain. Healthy blood vessels are able to widen and narrow in response to what the body needs at any given moment; high doses of caffeine can temporarily stop them from doing this, leaving them stiff and less responsive (Awashra et al., 2025). Grasser et al. (2014) demonstrated this effect directly, finding that Red Bull consumption led to increased resistance in the brain's blood vessels and a measurable slowdown in blood flow to the brain — even as the heart was beating harder and pumping more blood overall. This means that despite the heart working at a higher rate, the brain was actually receiving less blood than it would under normal conditions, which explains why some people feel dizzy or lightheaded after a strong energy drink. If this pattern is repeated regularly over months or years, the cumulative stiffening of blood vessels can become more permanent, turning a short-term chemical effect into a long-term structural problem for the heart and circulation. This therefore highlights that the harm from caffeinated drinks extends beyond the heart itself, affecting the entire network of vessels that keep the body supplied with blood and oxygen.

Several practical measures can be taken to reduce these harmful effects, both in the moment and as longer-term habits. In the short term, drinking water is one of the most effective responses, as caffeine removes fluid from the body and less fluid in the blood means more strain on the heart; rehydrating helps restore normal blood volume and assists the body in clearing the caffeine faster (Awashra et al., 2025). If a person is also sweating, shaking, or

feeling nauseous, replacing electrolytes — the minerals that help the heart maintain a steady rhythm — through a sports drink or coconut water can further stabilise the situation. Stopping any physical exercise immediately is also important, as exerting the body while the heart is already stressed by caffeine compounds the risk significantly. Over the longer term, Awashra et al. (2025) note that habitually consuming smaller, more spread-out amounts of caffeine allows the body to build a tolerance, which weakens the acute blood pressure response and makes the overall effects less severe. Equally important is learning to read labels carefully, as many energy drinks contain additional stimulants such as guarana whose caffeine content is not always included in the total shown on the packaging, and mixing energy drinks with alcohol — which masks how stimulated the body actually is — should always be avoided. Together, these strategies represent a realistic and evidence-based approach to managing the risks that caffeinated drinks pose to heart health in young adults.

References

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